HIRONAKA'S EXAMPLE OF A DESCENT DATUM ON A PROJECTIVE SCHEME OVER AN ETALE COVER THAT IS NOT EFFECTIVE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file examples.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: Hironaka example; companion of lemma-non-descending-property-projective.

Editorial note. The label lemma-non-effective-descent-projective names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. Revision r002: regenerated from the same spec after the r001 assembly failed at compile time (the author's citation of Hironaka via the bare key H had no bibliography backend in r001; the stale r001 job directory is retained as evidence).

Editorial note. Bibliography support: the counted proof cites the letter-key H (the Stacks Project's global bibliography entry for Hartshorne, Algebraic Geometry, Graduate Texts in Mathematics 52, Springer, 1977, whose Appendix B Example B.3.4.1/B.3.4.2 are the Hironaka examples being imitated). A matching hironaka.bib asset is attached so the author's own citation resolves; no statement or proof text was altered.

Allowed standard prerequisites (editorial).

- Groupoids, Definition groupoids-definition-principal-homogeneous-space: the notion of a G-torsor
- The construction and equivalence conclusion of lemma-non-descending-property-projective: the blowup tower over an open subset of projective three-space yields a proper non-projective morphism, and on the inverse image E of C union D there is a rational equivalence L-zero plus g of L-zero equivalent to zero
- Descent, Lemma descent-lemma-affine, noting that a closed immersion is an affine morphism: closed subschemes satisfy descent
- More on Algebra, Lemma more-algebra-lemma-regular-local-UFD: a regular local ring is factorial; hence on a smooth complex threefold an effective divisor gives rise to a line bundle
- Chow Homology, Lemmas chow-lemma-proper-pushforward-rational-equivalence and chow-lemma-factors: the degree of a line bundle on a proper scheme over a field is well defined on one-cycles modulo rational equivalence

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Date: stacks-project:examples.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 02TI: sources/stacks-chow/tags.txt, source lines 3697–3697. Prerequisite pointer invoked at 5583; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 02S2: sources/stacks-chow/tags.txt, source lines 3645–3645. Prerequisite pointer invoked at 5582; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0245: sources/stacks-chow/tags.txt, source lines 2766–2766. Prerequisite pointer invoked at 5556; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original definition 049A: sources/stacks-chow/tags.txt, source lines 5573–5573. Prerequisite pointer invoked at 5483; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 08J1: sources/stacks-examples/pinned-original-source.tex, source lines 5371–5371. Prerequisite pointer invoked at 5507; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0AG0: sources/stacks-chow/tags.txt, source lines 13178–13178. Prerequisite pointer invoked at 5574; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *There is an etale covering $X \to S$ of schemes and a descent datum $(V/X, \varphi)$ relative to $X \to S$ such that $V \to X$ is projective, but the descent datum is not effective in the category of schemes.*

Proof. We imitate Hironaka's example of a smooth separated complex algebraic space of dimension 3 which is not a scheme [1, Example B.3.4.2].

Consider the action of the group $G = \mathbf{Z}/2 = \{1, g\}$ on projective 3-space $\mathbf{P}^3$ over the complex numbers by

$$g[x, y, z, w] = [y, x, w, z].$$

The action is free outside the two disjoint lines $L_1 = \{[x, x, z, z]\}$ and $L_2 = \{[x, -x, z, -z]\}$ in $\mathbf{P}^3$. Let $Y = \mathbf{P}^3 - (L_1 \cup L_2)$. There is a smooth quasi-projective scheme $S = Y/G$ over $\mathbf{C}$ such that $Y \to S$ is a G-torsor (Groupoids, Definition 049A). Explicitly, we can define S as the image of the open subset Y in $\mathbf{P}^3$ under the morphism

$$\begin{aligned} \mathbf{P}^3 &\to \text{Proj } \mathbf{C}[x, y, z, w]^G \\ &= \text{Proj } \mathbf{C}[u_0, u_1, v_0, v_1, v_2]/(v_0 v_1 = v_2^2), \end{aligned}$$

where $u_0 = x + y$, $u_1 = z + w$, $v_0 = (x - y)^2$, $v_1 = (z - w)^2$, and $v_2 = (x - y)(z - w)$, and the ring is graded with u_0, u_1 in degree 1 and v_0, v_1, v_2 in degree 2.

Let $C = \{[x, y, z, w] : xy = z^2, w = 0\}$ and $D = \{[x, y, z, w] : xy = w^2, z = 0\}$. These are smooth conic curves in $\mathbf{P}^3$, contained in the G-invariant open subset Y, with $g(C) = D$. Also, $C \cap D$ consists of the two points $P := [1, 0, 0, 0]$ and $Q := [0, 1, 0, 0]$, and these two points are switched by the action of G.

Let $V_Y \to Y$ be the scheme which over $Y - P$ is defined by blowing up D and then the strict transform of C, and over $Y - Q$ is defined by blowing up C and then the strict transform of D. (This is the same construction as in the proof of Lemma 08J1, except that Y here denotes an open subset of $\mathbf{P}^3$ rather than all of $\mathbf{P}^3$.) Then the action of G on Y lifts to an action of G on V_Y, which switches the inverse images of $Y - P$ and $Y - Q$. This action of G on V_Y gives a descent datum $(V_Y/Y, \varphi_Y)$ on V_Y relative to the G-torsor $Y \to S$. The morphism $V_Y \to Y$ is proper but not projective, as shown in the proof of Lemma 08J1.

Let X be the disjoint union of the open subsets $Y - P$ and $Y - Q$; then we have surjective etale morphisms $X \to Y \to S$. Let V be the pullback of $V_Y \to Y$ to X; then the morphism $V \to X$ is projective, since $V_Y \to Y$ is a blowup over each of the open subsets $Y - P$ and $Y - Q$. Moreover, the descent datum $(V_Y/Y, \varphi_Y)$ pulls back to a descent datum $(V/X, \varphi)$ relative to the etale covering $X \to S$.

Suppose that this descent datum is effective in the category of schemes. That is, there is a scheme $U \to S$ which pulls back to the morphism $V \to X$ together with its descent datum. Then U would be the quotient of V_Y by its G-action.

$$\begin{array}{ccc} V & \longrightarrow & X \\ \downarrow & & \downarrow \\ V_Y & \longrightarrow & Y \\ \downarrow & & \downarrow \\ U & \longrightarrow & S \end{array}$$

Let E be the inverse image of $C \cup D \subset Y$ in V_Y; thus $E \to C \cup D$ is a proper morphism, with fibers isomorphic to $\mathbf{P}^1$ over $(C \cup D) - \{P, Q\}$. The inverse image of P in E is a union of two lines L_0 and M_0. It follows that the inverse image of $Q = g(P)$ in E is the union of two lines $L_0' = g(M_0)$ and $M_0' = g(L_0)$. As shown in the proof of Lemma 08J1, we have a rational equivalence $L_0 + M_0' = L_0 + g(L_0) \sim 0$ on E.

By descent of closed subschemes, there is a curve $L_1 \subset U$ (isomorphic to $\mathbf{P}^1$) whose inverse image in V_Y is $L_0 \cup g(L_0)$. (Use Descent, Lemma 0245, noting that a closed immersion is an affine morphism.) Let R be a complex point of L_1. Since we assumed that U is a scheme, we can choose a function f in the local ring $O_{U,R}$ that vanishes at R but not on the whole curve L_1. Let D_{loc} be an irreducible component of the closed subset $\{f = 0\}$ in $\operatorname{Spec} O_{U,R}$; then D_{loc} has codimension 1. The closure of D_{loc} in U is an irreducible divisor D_U in U which contains the point R but not the whole curve L_1. The inverse image of D_U in V_Y is an effective divisor D which intersects $L_0 \cup g(L_0)$ but does not contain either curve L_0 or $g(L_0)$.

Since the complex 3-fold V_Y is smooth, $O(D)$ is a line bundle on V_Y. We use here that a regular local ring is factorial, or in other words is a UFD, see More on Algebra, Lemma 0AG0. The restriction of $O(D)$ to the proper surface $E \subset V_Y$ is a line bundle which has positive degree on the 1-cycle $L_0 + g(L_0)$, by our information on D. Since $L_0 + g(L_0) \sim 0$ on E, this contradicts that the degree of a line bundle is well-defined on 1-cycles modulo rational equivalence on a proper scheme over a field (Chow Homology, Lemma 02S2 and Lemma 02TI). Therefore the descent datum $(V/X, \varphi)$ is in fact not effective; that is, U does not exist as a scheme. □

References

[1] Robin Hartshorne. *Algebraic Geometry*. Number 52 in Graduate Texts in Mathematics. Springer-Verlag, New York-Heidelberg, 1977.